German International University

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MECHANICAL ENGINEERING

livox Lidar Point Cloud Segmentation and Object Detection based on Artificial Neural Networks

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Submission Date: 30/01/2024

Acknowledgments

I extend my deepest gratitude to my advisor, Prof. Dr. rer. nat. Toralf Trautmann, for their invaluable guidance and support throughout my research journey. Their expertise and insights have been fundamental to the success of this project. I am also grateful to Mr. Ammad for his contributions and the unique perspective he brought to this work. My sincere thanks go to my lab team as well; their collaboration and assistance have been instrumental in the progress and completion of this study.

Abstract

LiDAR sensors are knowm for their ability to generate detailed three-dimensional maps, providing a dataset that is quite different from the two-dimensional images typically utilized in object detection.

The thesis begins by collecting LiDAR data in LVX format, which is unique to Livox sensors. This data is then converted to a more manageable Point Cloud Data (PCD) format, facilitating effective processing and analysis. A significant challenge addressed in the thesis is adapting the YOLO v4 algorithm, which is originally designed for two-dimensional image data, to effectively work with three-dimensional LiDAR data. This adaptation includes data preprocessing approaches and accurate data labeling to ensure accurate training inputs.

The training of the neural network requires fine-tuning a variety of parameters to match the unique characteristics of the LiDAR data. The effectiveness of the training is quantitatively assessed using the Intersection Over Union (IOU) method, which measures the model's precision in identifying and localizing objects by comparing the tested data with labeled data.

A comprehensive literature review is included, encompassing both the technical and practical aspects of training the YOLO v4 model. The study's findings, which detail the model's functionality and potential applications, are presented in the results section. The research aims to contribute to the field of autonomous systems and improve the integration of complex neural networks with LiDAR technology for precise object detection

Contents

Chapter 1

Introduction

Figure 1.1: Pre data collection.

Figure 1.2: Installing sensors.

In this thesis, we explore the intersection of advanced machine learning and LiDAR technology, specifically focusing on training a neural network, YOLO v4, to detect objects using data from a Livox LiDAR sensor. LiDAR sensors, known for their ability to create detailed three-dimensional maps of their surroundings, offer a rich dataset that is significantly different from the traditional two-dimensional images typically used in object detection.

first of all it begins with the collection of LiDAR data in LVX format, a unique format specific to Livox sensors. This format provides a comprehensive representation of the environment but requires conversion to a more usable Point Cloud Data (PCD) format so it becomes easy to deal with. This conversion is crucial for our objective, as it prepares the data for effective processing and analysis.

The core challenge addressed in this thesis is adapting the YOLO v4 algorithm, originally designed for two-dimensional image data, to work effectively with three-dimensional LiDAR data. This involves an innovative approach to data pre processing, including the transformation of 3D data into a format suitable for the YOLO v4 model, and accurately labeling the data to provide accurate training inputs.

Training the neural network involves fine-tuning various parameters and settings to accommodate the unique characteristics of the LiDAR data, followed by the training process. The effectiveness of this training is quantitatively evaluated using the Intersection Over Union (IOU) method, which provides a measure of the model's accuracy in identifying and localizing objects within the data by comparing the tested data with labelled data .

This thesis covers a wide range of literature review in addition to the technical and practical aspects of training the YOLO v4 model, putting our investigation into the context of continuing fieldwork. The results of our training process are summarized in the results section, which offers details on the model's functionality and potential uses..

Through this research, we aim to contribute to the growing field of autonomous systems and enhance our understanding of integrating complex neural networks with LiDAR technology for accurate object detection

Chapter 2

Advancement in livox lidar sensor and neural network applications

2.1 overview of LIDAR sensor

Automotive LiDAR (Light Detection and Ranging) is a critical technology in the development of autonomous vehicles, offering unique environmental data unattainable by other sensors. LiDAR (Light Detection and Ranging) is a important technology in autonomous vehicles, offering unique environmental data that other sensors cannot capture. Its core mechanism, Time-of-Flight (ToF), calculates distance by measuring the travel time of emitted light, represented by the equation:

$$R = \frac{c\Delta T}{2} \tag{2.1}$$

where R is the range in meters, c is the speed of light in meters per second, and ΔT is the round-trip time in seconds.

Among the various LiDAR systems, scanning LiDAR with micro electromechanical mirror systems (MEMS) stands out for its potential to reduce size and cost while maintaining high performance.

The receiver chain in these systems, generally comprising photo diodes and transimpedance amplifiers, significantly influences the amount of data collected. This data volume, often reaching several gigabits per second, presents substantial challenges for real-time processing in vehicle systems, especially when integrating multiple LiDAR sensors. The subsequent sections explore key LiDAR parameters affecting data volume and examine strategies for managing this high data rate. . Among various LiDAR systems, scanning LiDAR with micro electromechanical mirror systems (MEMS) is noted for its potential in reducing size and cost while maintaining high performance.

The receiver chain of these systems, typically comprising photo diodes and transimpedance amplifiers, significantly impacts the volume of data captured. This data, often amounting to several gigabits per second, poses a substantial challenge for real-time processing in vehicle systems, especially when integrating multiple LiDAR sensors. The paper proceeds to explore key LiDAR parameters affecting data volume, strategies for data rate reduction, and concludes with an overview of these findings . Networks [1] .

2.2 LIVOX LIDAR SENSOR

LIVOX HORIZON is the LIDAR sensor chosen for the project as it has a wide range of applications due to its advanced features , capabilities and and the catalogue of LIVOX HORIZON [2] describe its specifications .

2.2.1 technical specifications of LIVOX HORIZON

Technical Specifications of LIVOX HORIZON are

- Laser Wavelength: 905 nm.
- Laser Safety: Classified as Class 1 according to IEC 60825-1:2014, ensuring it is safe for eyes.
- Detection Range: 90 m at 10 percent of reflectivity. 130 m at 20 percent of reflectivity. 260 m at 80 percent of reflectivity.
- Field of View (FOV): 81.7° Horizontal x 25.1° Vertical.
- Distance Random Error: Less than 2 cm at 20 m.
- Angular Random Error: Less than 0.05º.
- Beam Divergence: 0.28° Horizontal x 0.03° Vertical.
- Point Rate: 240,000 points/s (first or strongest return). 480,000 points/s (dual return).
- False Alarm Ratio: Less than 0.01 percent at 100 klx

2.2.2 Applications of LIVOX HORIZON

- Autonomous Vehicles: Its precise and detailed environmental mapping is crucial for autonomous navigation in cars, drones, and other autonomous systems.
- Robotics: In robotics, the Livox Horizon's accurate distance measurements and 3D mapping capabilities are essential for navigation, obstacle avoidance, and environment interaction.

- Surveying and Mapping: It is used in aerial surveying, topographical mapping, and urban planning due to its ability to generate detailed 3D point clouds of large areas.
- Security and Surveillance: The sensor can be employed in security systems for area monitoring, crowd management, and intrusion detection.
- Agriculture: In precision agriculture, it helps in crop monitoring, field analysis, and resource management

2.2.3 Addvantages and of LIVOX HORIZON

- Non-Repetitive Scanning Technology: This unique technology, along with multi-laser and multi-APD DL-Pack technology, ensures a high-density point cloud that is approximately three times denser than the Livox Mid series within the same period.
- Detailed Environment Representation: Over time, the coverage inside the FOV increases significantly, revealing more detailed information about the surroundings .

2.3 YOLOv4 in Object Detection

IN my project i have followed the methodology provided by my mathworks [3]

YOLOv4 YOLOv4 (You Only Look Once version 4) is a significant improvement in real-time object detection models, particularly excelling in terms of inference speed. It incorporates a variety of computer vision techniques along with a few novel contributions **roboflow˙yolov4**.

This research [4] focuses on optimizing YOLOv4 for practical use in real-world applications. its methodology includes several key innovations:

- Architecture: YOLOv4 utilizes CSPDarknet53 as the backbone, with SPP and PAN used for additional network enhancements. This combination allows for efficient processing and accurate object detection.
- Data Augmentation: Techniques like Mosaic and Self-Adversarial Training (SAT) are used to improve the robustness of the model. Mosaic combines four different training images to enhance the context understanding of the model, while SAT involves two stages of training to improve detection accuracy.
- Optimization Techniques: The study employs various optimization techniques like using different types of losses (CIoU, DIoU) for bounding box regression, which help in improving the precision of object localization.

- Bag of Freebies and Specials: They incorporate 'Bag of Freebies' (BoF) and 'Bag of Specials' (BoS) methods to enhance training and inference without significantly increasing computational cost. These include various activation functions, normalization methods, and regularization techniques.
- Efficient Training: The model is designed for efficient training on single GPUs, making it accessible for wider usage without the need for specialized hardware.

also this research [5] discusses an improved version of YOLOv4, called YOLOv4-tiny, tailored for real-time object detection on embedded devices. The paper proposes modifications to simplify the network structure and reduce computational complexity, while maintaining detection accuracy and here is the methodology of this paper

- Network Architecture: YOLO V4 employs a specific architecture tailored for efficient object detection. This includes the use of convolutional layers, batch normalization, and leaky ReLU activation functions.
- Training: The model is trained on a large dataset with annotated images, using a loss function that accounts for object classification, localization, and confidence in detections.
- Detection Process: In the detection phase, YOLO V4 processes an entire image in a single pass and predicts bounding boxes and class probabilities for these boxes.
- Post-processing: After initial predictions, post-processing steps like non-maximum suppression are used to refine the results, eliminating redundant and overlapping boxes.

YOLO v4 is used in detection of small objects , This paper [6] focuses on enhancing the YOLO V4 algorithm for improved detection of small objects. It introduces a novel network structure featuring multi-scale contextual information, aiming to extract more effective features from small objects the methodology of this paper goes like this

- Enhanced Network Structure: The YOLO V4 architecture incorporates a feature-enhanced network based on multi-scale contextual information. This network is optimized to extract more discriminative and robust features from small objects.
- Soft-CIOU Loss Function: The Soft-CIOU loss function is proposed to improve the learning capability of the network for small objects. It introduces a square root operation on the Euclidean distance and an aspect ratio weighting factor to the Complete-IOU (CIOU) loss function.
- Attention Module Integration: To minimize the influence of insignificant information on object detection, an attention module is introduced. This module focuses on channels with small objects during upsampling and pinpoints regions with objects during downsampling.

- Experimental Validation: The effectiveness of these enhancements is validated through experiments on publicly available small object datasets. The results demonstrate better detection performance for small objects compared to other models.
-

2.4 point cloud in object detection

2.4.1 introduction to point cloud

A point cloud is a set of data points in space, commonly used in the field of computer vision and graphics. These points represent the external surface of an object or a scene, captured by devices like LiDAR scanners or 3D scanners. Each point in the cloud is represented by its coordinates in a three-dimensional space, and sometimes additional information like color or intensity , This paper "Deep Learning for 3D Point Clouds [7] discusses the importance of point clouds in deep learning for 3D object recognition and scene understanding. It highlights the challenges and recent advancements in applying deep learning techniques to point cloud data The document "Point-cloud-based 3D object detection and classification methods for self-driving applications: A survey and taxonomy" [8] provides a comprehensive analysis of point cloud-based methods in 3D object detection, especially for self-driving applications. It discusses various approaches to data representation (like point-based, voxel-based, frustum-based, pillar-based, projection-based), feature extraction methods, and the architecture of detection networks. The paper insures the challenges of processing point clouds due to their high-dimensional and limited nature and explores different techniques for efficient data representation and feature extraction. It also highlights the importance of integrating various sensors and data types for more effective object detection in autonomous vehicles.

This [9] research paper "Learning Representations and Generative Models for 3D Point Clouds" emphasizes the importance of point clouds in deep learning, particularly for 3D object representation. It introduces advanced techniques in encoding and generating point clouds using deep learning architectures, such as autoencoders and generative adversarial networks (GANs). The study highlights the challenges in processing point clouds due to their unstructured nature and proposes innovative solutions for effective representation learning and generative modeling. This research is significant in enhancing the capabilities of deep learning models to handle complex 3D data, beneficial in fields like robotics, virtual reality, and autonomous vehicles.

2.4.2 point clouds in object detection

The methodology for object detection using point clouds in the paper "Point-cloud-based 3D Object Detection and Classification Method for Autonomous Driving"

[8] involves several steps:

- Data Acquisition and Preprocessing: The study uses LiDAR sensors to acquire point cloud data. The raw data is preprocessed to filter out noise and irrelevant information, enhancing data quality for detection tasks.
- Feature Extraction: The method extracts features from the preprocessed point cloud data. This includes identifying key points and descriptors that are crucial for recognizing different objects in the data.
- Object Detection and Classification: Using the extracted features, the system detects and classifies objects within the point cloud. This step often involves using machine learning algorithms or neural networks tailored for 3D data interpretation.
- Post-processing: After the initial detection and classification, post-processing steps such as bounding box refinement and object tracking are performed to improve the accuracy and reliability of the results.
- Evaluation: The methodology includes rigorous evaluation of the detection system, usually involving metrics like precision, recall, and Intersection Over Union (IOU) to measure performance.

2.5 LIDAR in object detection

Lidar is a remote sensing technology that measures distances by illuminating a object with laser light and analyzing the reflected light [10] .

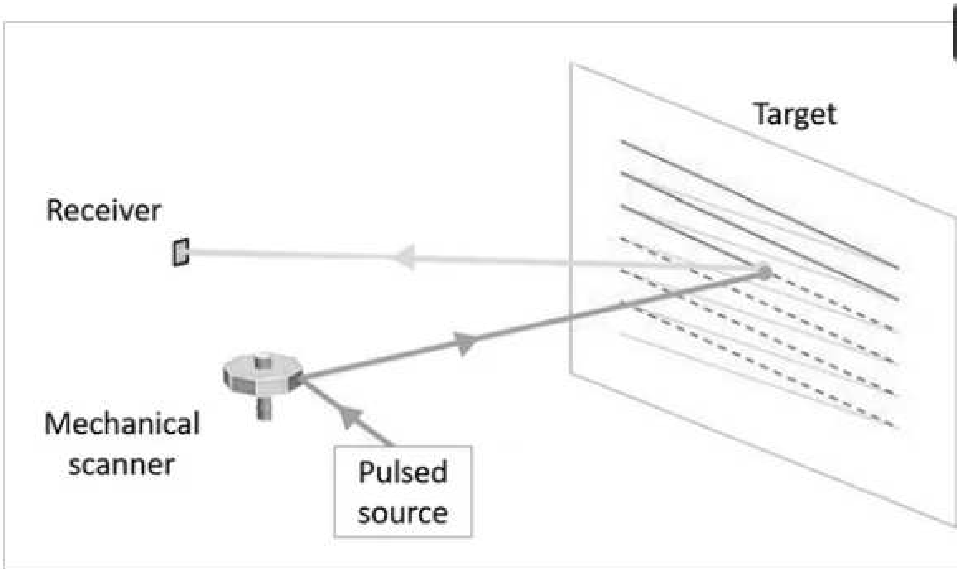

Figure 2.1: Illustration depicting a standard mechanical scanning-based Light Detection and Ranging (LiDAR) imaging system configuration

[11]

LIDAR sensors are commonly used in 3d object detection . this resaech [12] shows how LIDAR sensors are used in object detection and it gives a comprehensive details about using LIDAR sensors in object detection the methodolgy of it is :

- LiDAR Sensing Techniques: It discusses LiDAR technology and its applications in 3D object detection, including different LiDAR ranging and scanning methods.
- Deep 3D Detection Networks: The paper categorizes deep learning networks for LiDAR data based on point cloud representations and explores their challenges.
- Object Detection Network Structure: Various network structures for 3D object detection are reviewed, including their input and output encoding methodologies.
- Evaluation Metrics and Dataset Analysis: It summarizes the evaluation metrics and performance analysis of algorithms on authoritative 3D detection benchmarks.
- Methodological Contributions and Insights: The paper offers insights into the challenges and future directions in the field, providing a comprehensive understanding of LiDAR-based 3D object detection.

Also other research like [13] has obtained how the LIDAR sensor in object detection which the the methodology for it is :

-
- Bird's Eye View (BEV) Generation: LiDAR point cloud data is converted into a BEV image. This image encodes information like height, intensity, and density of points.
- Density Normalization: To account for variations in LiDAR sensors, a normalization process is applied to the BEV image, ensuring consistent and meaningful values.
- Object Detection Using Faster R-CNN: The BEV images are processed using the Faster R-CNN architecture, adapted to handle these 2D structures. This step involves detecting object locations and orientations.
- Multi-Task Training: The model is trained using a multi-task loss function that includes object detection, classification, and orientation estimation.
- Post-Processing: The network's output is further processed to obtain accurate 3D object detections. This includes refining axis-aligned detections to object-aligned 2D boxes and estimating object height.
- Data Augmentation: Techniques like horizontal flipping and rotation of annotations are used to improve the robustness and accuracy of the model.

so there are many applications using LIDAR sensor . [14]

Chapter 3

methodology

3.1 Data Collection

The Livox LiDAR sensor, known for its high precision and wide field of view [15], was employed to collect data under six scenarios:

- A pedestrian walking along a path.

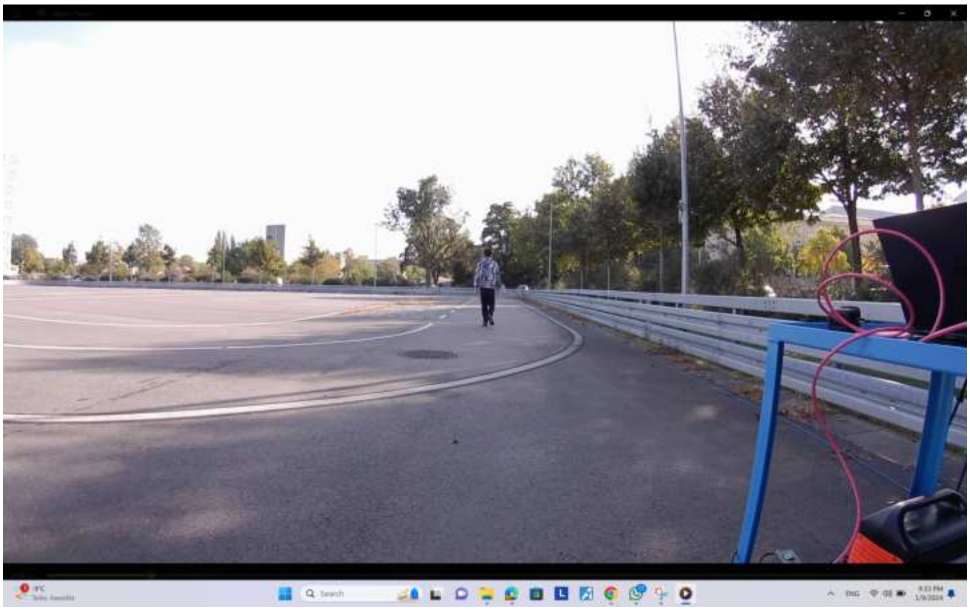

Figure 3.1: in this scenario a single pedestrian walks straight in a long path then back again straight

- Multiple pedestrians to simulate crowd dynamics.

Figure 3.2: in this scenario three pedestrians walk randomly by each other

.

- A pedestrian changing lanes.

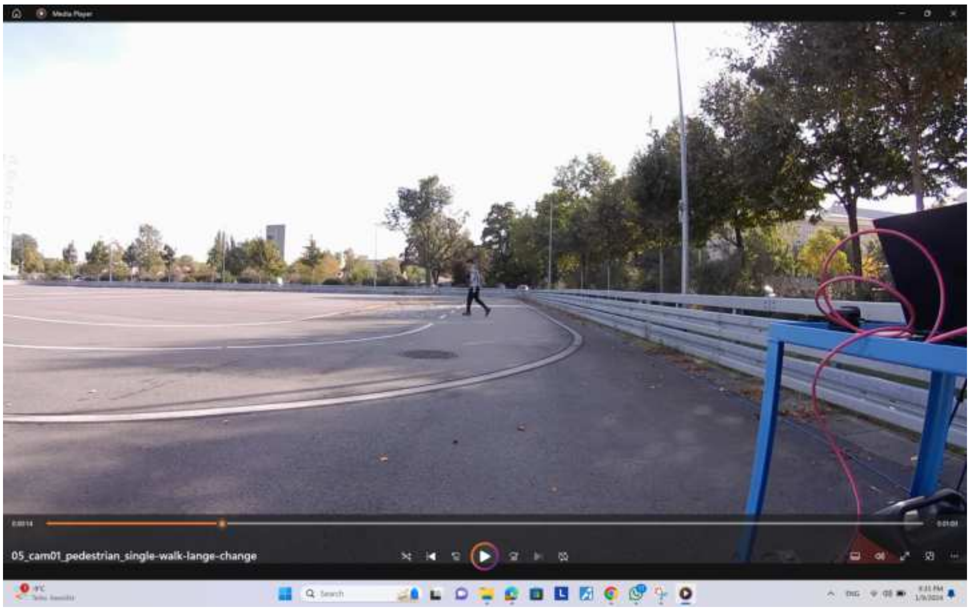

Figure 3.3: in this scenario a single pedestrian is walking changing his lane randomly by each other

- Cars Lane Change. in This scenario vehicles engaging in lane-switching activities, providing a glimpse into the intricate dynamics of driving behavior
- Cars Cut Out. in this scenario vehicles performing various maneuvers, like swiftly changing lanes, the model is evaluated on its reaction to unexpected alterations in trajectory.

- Cars Single Lane in this scenario Vehicles are captured in moving inside a single lane which represents a typical traffic flow in controlled driving conditions.

3.2 Data Conversion

The labelled data was collected in LVX format which LIVOX can deal with . Data in LVX format were converted to PCD format for compatibility with processing tools. A specialized Python script was developed for this purpose, which includes functions to read LVX files, extract relevant data, and reformat them into PCD format.

3.3 labelling

labelling is done via ground truth labeler app in Matlab [16] ,

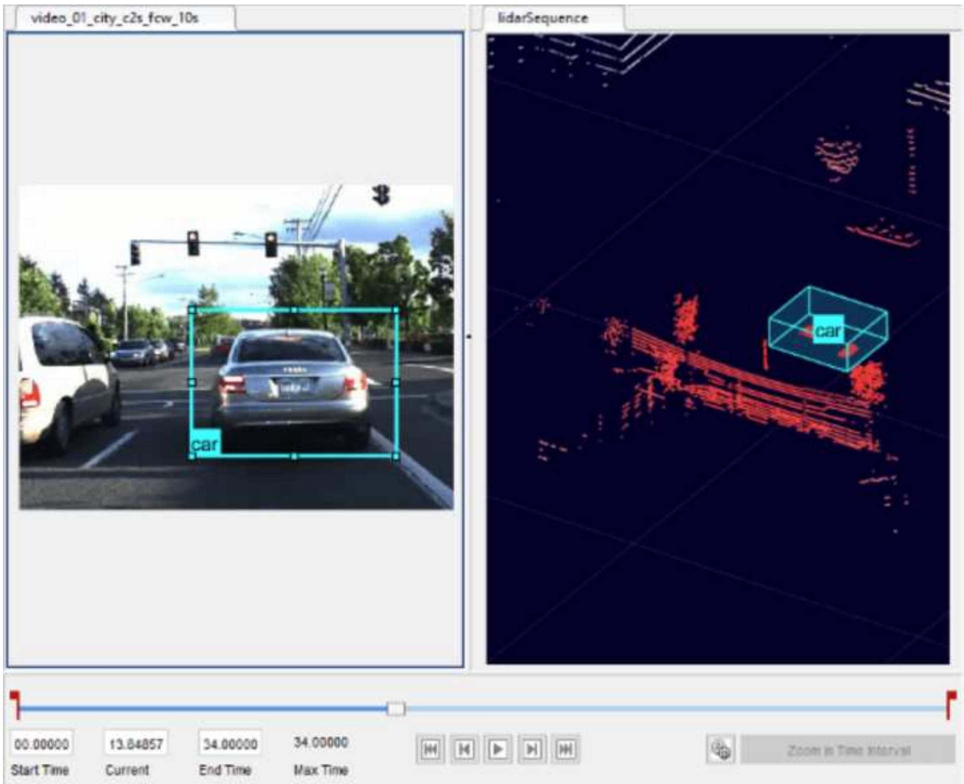

Figure 3.4: Gtruth labeler

in each frame of the six scenarios labelling is conducted by drawing bounding boxes around the objects manually and labeling them with the correct classes and this is done to train the network to do it by itself .

In this scenario a bounding box is conducted around the object which is here a car and point clouds of the car is properly indicated using projected view tool which shows the front , side and top view which makes it easy for labelling.

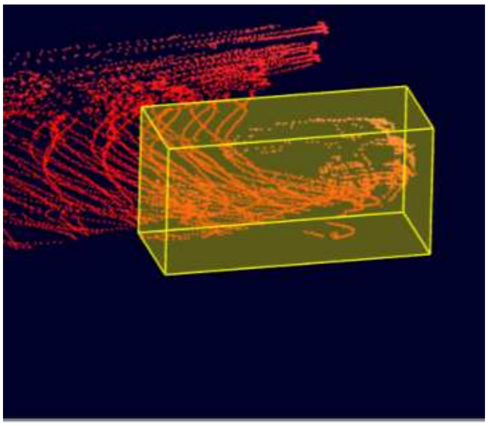

Figure 3.5: point clouds of a car is captured in this figure .

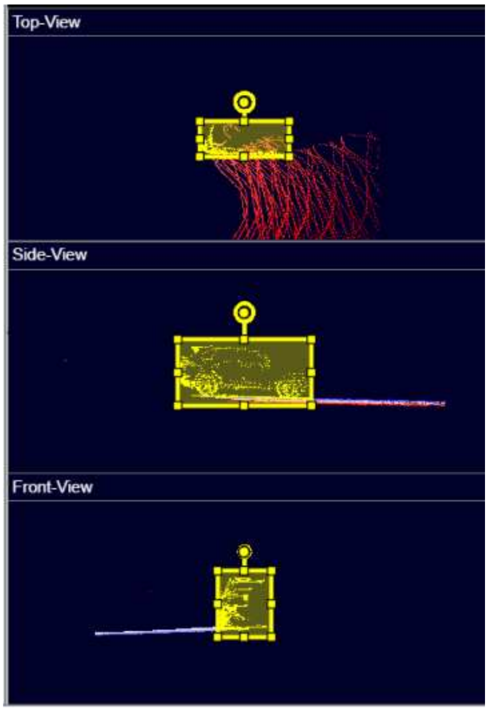

Figure 3.6: labelling was done using projected view feature.

in some scenarios there were some problem indicating the pedestrian as there are no point clouds for the pedestrian .

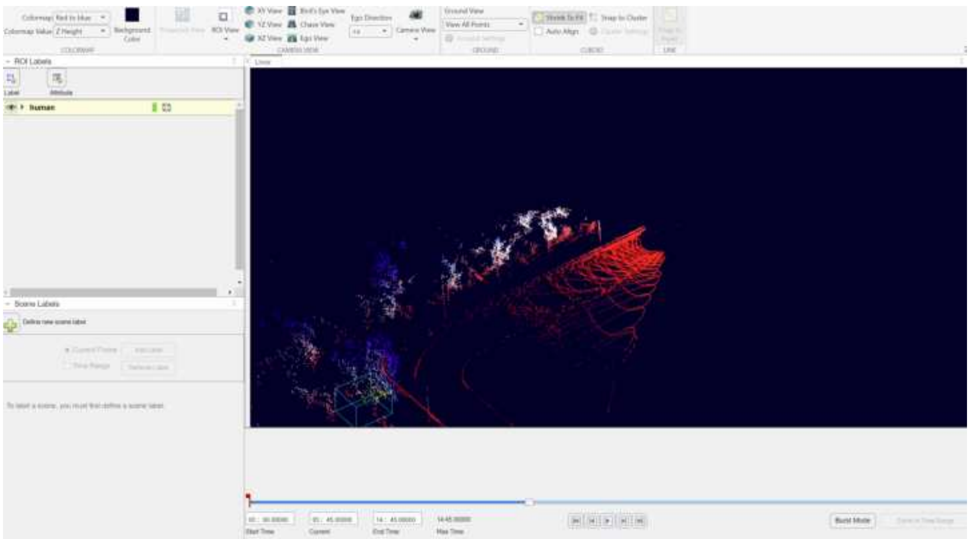

Figure 3.7: problem indicating the point clouds of the object

3.3.1 G Truth

Once the labeling of all captured frames was completed, the resulting ground truth data was exported. This data is now primed for use in the neural network's training phase.

The gTruth function is explained as follows:

Data Sourcing : This sets the Data Sourcing property and identifies the source of the ground truth data. LabelDefs: This component outlines the details of labels, sublabels, and attributes, assigning them to the LabelDefinitions property. LabelData: This aspect handles the identification and positioning of marked labels, along with their timestamp data, forming the basis of the LabelData property. [17]

3.4 chossing model

there are two options for the network , first one is building a network from scratch and the other one is using a pretrained model the first one require a massive data and time so YOLOv4 was chosen for due to its high speed and efficiency, which makes it ideal for real-time applications. It strikes a balance between accuracy and speed, supports a wide range of object classes, and is relatively easier to train and implement compared to other models. This makes YOLOv4 a good choice for practical and commercial applications where real-time processing is crucial .

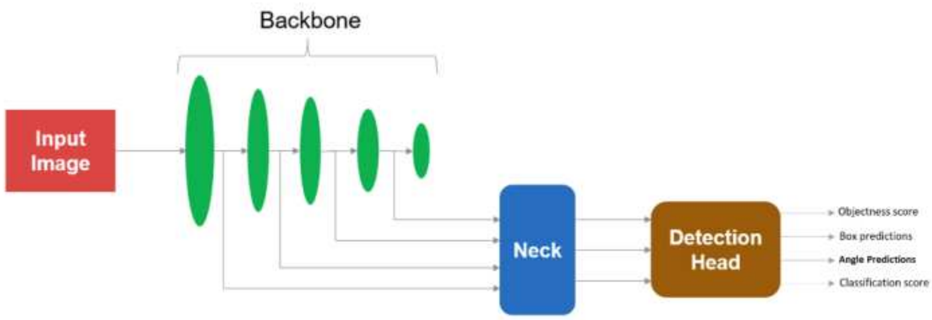

Figure 3.8: YOLO V4 ARCHITEACTURE

The above figure 3.8 represent the architecture of YOLO V4 described in this research [18] neural network as :

- Input Image: This is the raw image that is fed into the network for object detection.
- Backbone: The backbone is the part of the network responsible for extracting features from the input image. It usually consists of a pre-trained convolutional neural network (CNN) such as ResNet, Darknet, CSPNet, etc. The backbone processes the image through a series of convolutional layers to create a feature map.

- Neck: The neck is a series of additional layers and processing steps that sits between the backbone and the detection head. Its function is to refine the feature maps produced by the backbone, often aggregating features at different scales and enhancing the feature representation for more accurate detection. Techniques like Feature Pyramid Networks (FPN) or Path Aggregation Networks (PAN) are commonly used here.
- Detection Head: This is the final part of the network that makes the actual detection. It uses the processed feature maps to predict bounding boxes, object class probabilities, and other attributes such as objectness score, which indicates the likelihood of an object being present in a box

3.5 Network Training

The Complex-YOLO v4 network was customized for processing LiDAR data following these steps :

3.5.1 preparing data

Converting labeled 3D point cloud's into a 2D BEV format as YOLO V4 can deal with 2D images and Ensuring annotations (bounding boxes and labels) are correctly mapped in this conversion.

3.5.2 Dataset Splitting

Splitting data into training, validation, and testing sets.

3.5.3 Setting Training Parameters

in this step training parameters were used for training the network . the parameters are :

- maxEpochs: The number of complete passes through the entire dataset.
- miniBatchSize: The number of samples that will be processed before the model updates its weights.
- learning Rate: Determines the step size at each iteration while moving toward a minimum of the loss function.
- warm up Period: A number of initial iterations where the learning rate is gradually increased to the full value to stabilize training.

- l2Regularization: Also known as weight decay, this parameter helps to prevent the model from over fitting.
- penaltyThreshold: Often used in object detection to define the threshold for considering a detection to be a true positive or false positive.
- velocity: Used in optimization algorithms like SGD with momentum; it refers to the term that accumulates the gradient descent velocity for weight updates .

```
Rng(10) for both
maxEpochs = 50;
miniBatchSize = 16;
learningRate = 0.001;
warmupPeriod = 500;
l2Regularization = 0.001;
penaltyThreshold = 0.5;
velocity = [];
```

Figure 3.9: Final training options.

3.5.4 Evaluation

Using the validation set to evaluate the model during training by Adjusting hyper parameters .

3.5.5 Fine-Tuning and Optimization

Based on testing results, it is essential to fine-tune the model by adjusting hyper parameters or retraining with augmented data.
The final hyper parameters are shown in figure 3.9.

3.6 Intersection Over Union

Post-training, the model's performance was tested using the Intersection Over Union (IoU) metric. The IoU offers a robust method to gauge detection accuracy across various scenarios. Critically, it requires the conversion of the trained data back into a three-dimensional format to facilitate a direct comparison with the 3D labeled data, thus providing a solid quantitative foundation for comparative analysis [19]. The IoU is computed as follows:

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \tag{3.1}$$

as shown in figure 3.10

The `calculate3DIoU` function is engineered to process the centroid coordinates (X_{center}, Y_{center}, Z_{center}) and the dimensions (length, width, height) of a pair of 3D bounding boxes. The initial step involves calculating the volume of intersection between the two boxes, as outlined in the subsequent equation:

$$V_{\text{intersection}} = \max(0, \min(x_{max1}, x_{max2}) - \max(x_{min1}, x_{min2})) \times \max(0, \min(y_{max1}, y_{max2}) - \max(y_{min1}, y_{min2} \tag{3.2}$$

Following this, the function calculates the union volume of the two boxes, incorporating the intersection volume from the previous step:

$$V_{\text{union}} = (l_1 \cdot w_1 \cdot h_1) + (l_2 \cdot w_2 \cdot h_2) - V_{\text{intersection}} \tag{3.3}$$

The IoU metric for 3D bounding boxes is then determined by the ratio of the intersection volume to the union volume. This ratio quantifies the extent of overlap between the boxes and is mathematically represented as:

$$\text{IoU} = \frac{V_{\text{intersection}}}{V_{\text{union}}} \tag{3.4}$$

This IoU metric is instrumental in validating the accuracy and reliability of 3D object detection techniques, particularly in the context of point cloud data within computer vision applications.

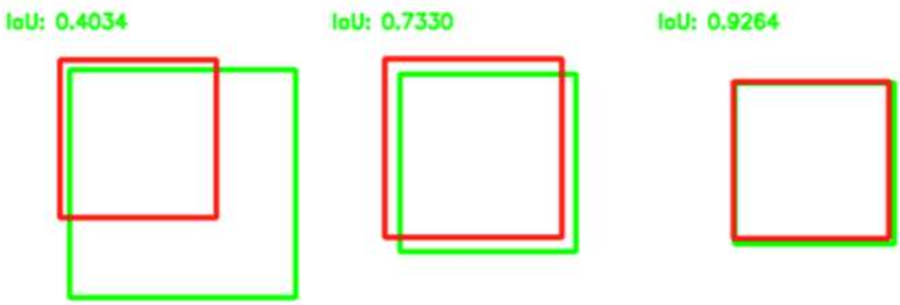

Figure 3.10: IOU method

Chapter 4

results

4.1 analysis of the results

	AOS	AP	OrientationSimilarity	Precision	Recall
Car	0.49568	0.49606	{368×1 double}	{368×1 double}	{368×1 double}
Pedestrain	0.24993	0.24998	{564×1 double}	{564×1 double}	{564×1 double}

Figure 4.1: AOS and AP results

The above figure 4.1 shows the results for both cars and pedestrian detection . AOS value for cars is better than AOS value for pedestrians so network could better detect cars better than pedestrians as AOS value for pedestrians is 0.24993 while for cars it is 0.49568.

[20]

The AP values are similar to the AOS values, with cars having an AP of 0.49606 and pedestrians having an AP of 0.24998. This indicates that the model's ability to correctly identify cars as cars is about twice as good as its ability to identify pedestrians.

From this table, we can deduce that the object detection system is more effective at detecting cars than pedestrians. The AOS and AP are significantly higher for cars, suggesting that the model is more reliable in its car detection. For pedestrians, the lower values indicate that there are incorrect detection or the model struggles with accurately determining their orientation.

4.2 learning rate and loss graph

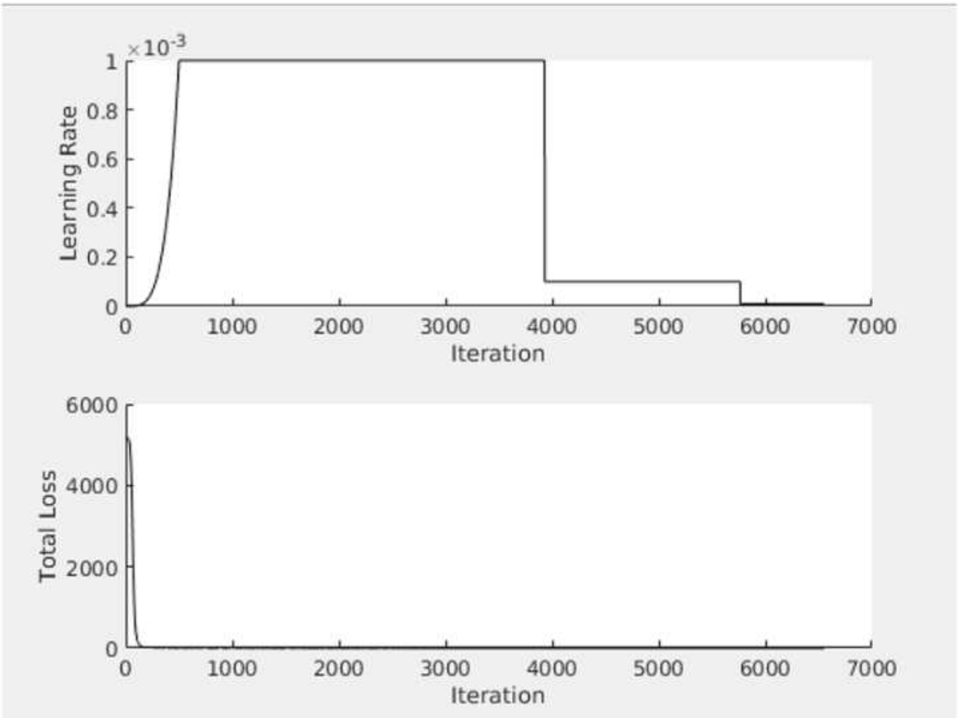

Figure 4.2: results of training model

The first graph shows the learning rate schedule in the beginning of training the network the learning rate increases gradually , then it takes a constant value and in the end of the iteration it starts to decrease This pattern is typical for stabilizing training initially and then refining the model with smaller updates as it converges.

in the second graph graph the loss is very enormous in the beginning of training the network after the iteration the loss begins to decrease The smooth decline indicates the model is learning effectively without significant instability or plateaus that could suggest problems in training.

4.3 Trained model detection

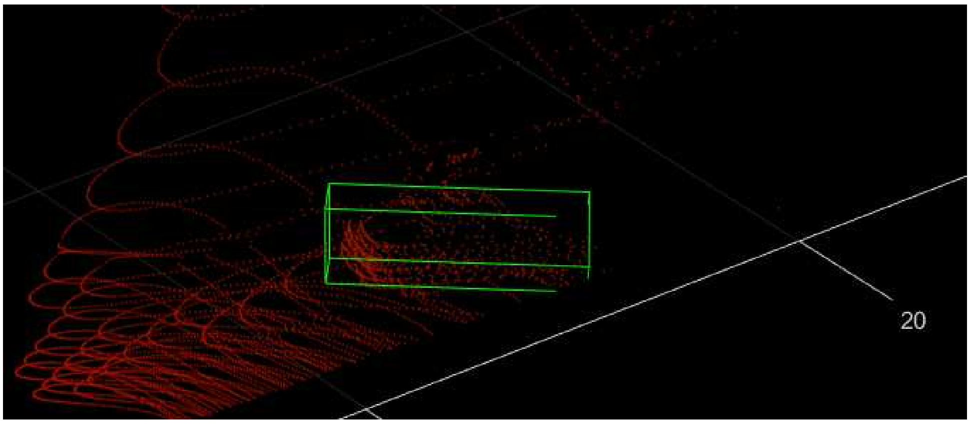

Figure 4.3: detected car using the trained model

as shown in the above figure 4.3 the model has predicted the car with the bounding box and this reflects the AOS values that shows that the model can detect 'car' class with a high accuracy .

4.4 IOU results

4.4.1 car IOU result

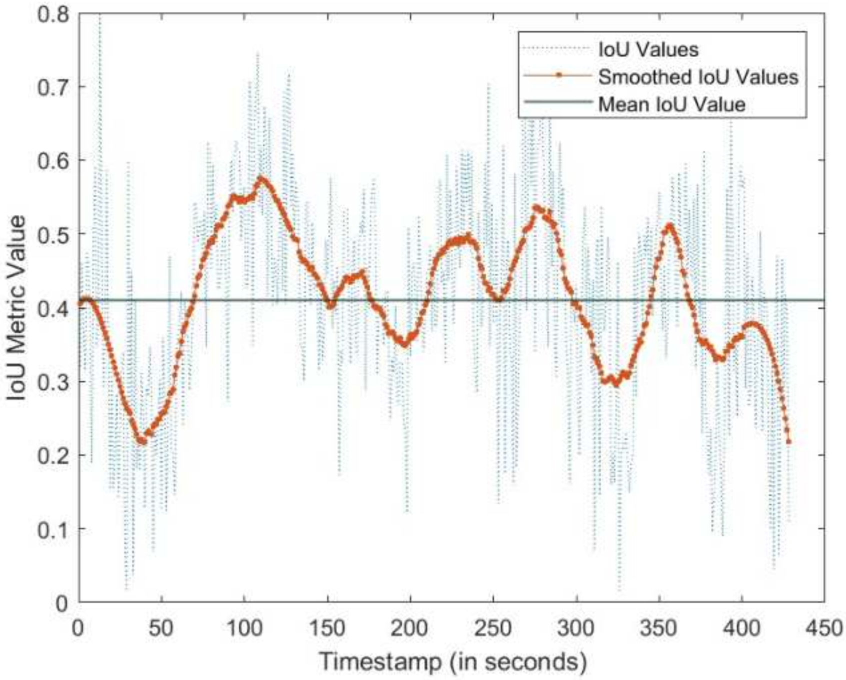

Figure 4.4: IOU result of the first scene

AS shown in figure 4.4 in the beginning when the objects were close to the sensor it was difficult to detect the car so the IOU results were in low ranges , when the objects are far from the sensor the network could better detect the objects so the average of detecting cars is 0.4 .

4.4.2 pedestrian IOU result

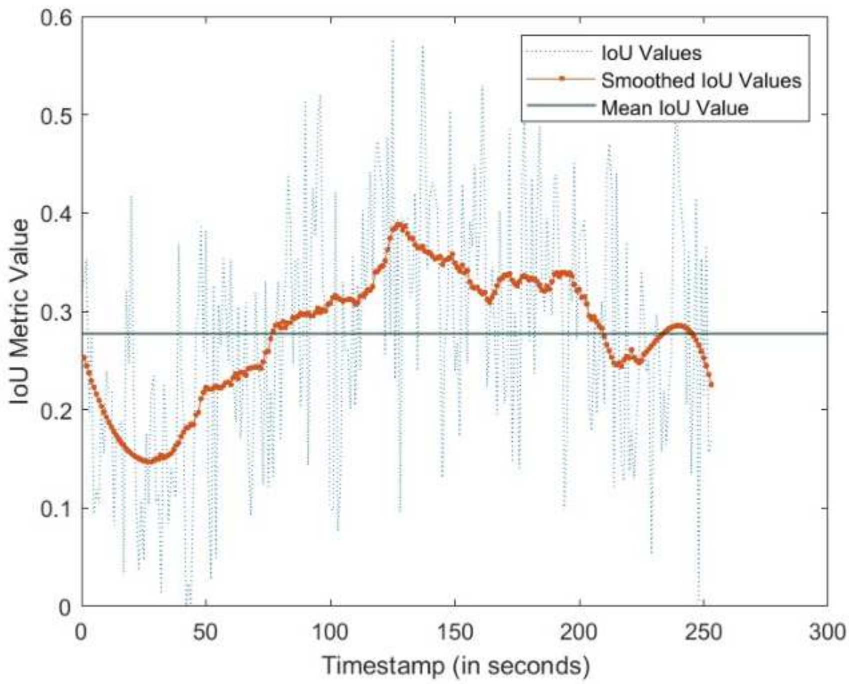

Figure 4.5: IOU result of the sixth scene

As shown in figure 4.5 in the begging it was difficult for the network to detect the pedestrians when they are close to the sensor when the pedestrians getting far from the sensor the network is doing much better in detecting the pedestrians and when the pedestrians are getting back again to the sensor the IOU metric value getting low again so the average of the IOU metric value is less than 0.3 which is lower than the metric value for the cars .

4.5 discussion of the results

The results indicate that the model is more accurate at detecting cars compared to pedestrians. This suggests that the model may benefit from additional training specifically targeted at pedestrian features to improve its detection capabilities for non-vehicular objects.

Chapter 5

Conclusion

utilizing the YOLO v4 neural network as a central technology, this thesis has explored the field of object detection utilizing LiDAR data. YOLO v4 is successfully modified for processing LiDAR point clouds during the course of the study, which is a notable departure from its typical application with 2D images. This modification required a thorough approach for gathering, converting, and labeling data to guarantee that the network was trained with accurate and pertinent information.

in the project the network could identify objects, especially cars, with a high degree of accuracy as indicated by the AOS and AP measures. Nevertheless, the network's ability to identify pedestrians was comparatively less effective, suggesting a possible area for future development and training the network on more data .

The study also demonstrated the difficulty and promise of combining cutting-edge neural networks with LiDAR technology, opening the door to more complex uses in autonomous driving and other fields needing accurate spatial awareness and object identification.

To sum up, this thesis emphasis the knowledge in autonomous system technologies by providing insights into the fusion of LiDAR data and neural networks for improved object detection capabilities.

Chapter 6

appendix

6.1 abbreviations and terms

- AOS: A combined metric that accounts for both the detection precision and the accuracy of the orientation estimation of bounding boxes.
- AP: Average Precision, which summarizes the precision-recall curve as the mean of precision achieved at each threshold.
- OrientationSimilarity: This refers to a measure that compares the orientation of the predicted bounding boxes to the ground truth.
- Precision: The ratio of true positive detections to the total number of positive detections (true positives + false positives). It reflects how many of the detections were actually correct.
- Recall: The ratio of true positive detections to the total number of actual positives (true positives + false negatives). It indicates how many of the actual objects were detected
- PCD : PCD stands for Point Cloud Data
- LIDAR : Light Detection and Ranging
- YOLO : you only look once .

6.2 List of figures

Figure 6.1: car lane change

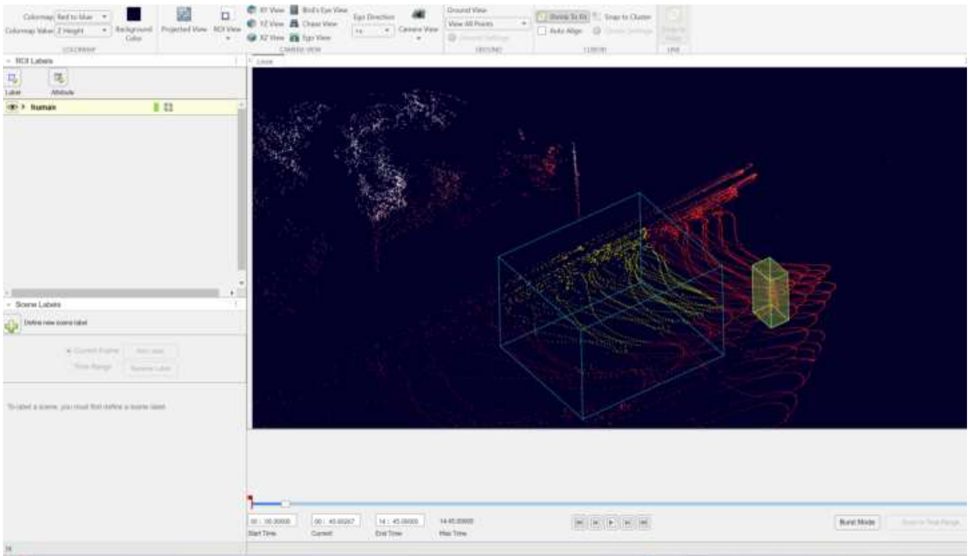

Figure 6.2: labelling pedestrians

Figure 6.3

Figure 6.4: two cars cut out

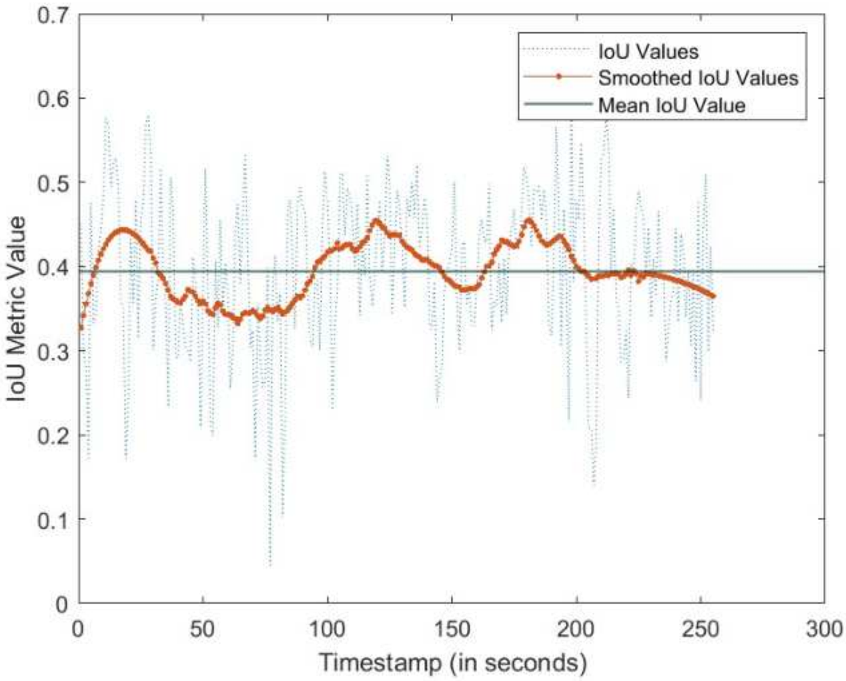

Figure 6.5: IOU graph for scene 3

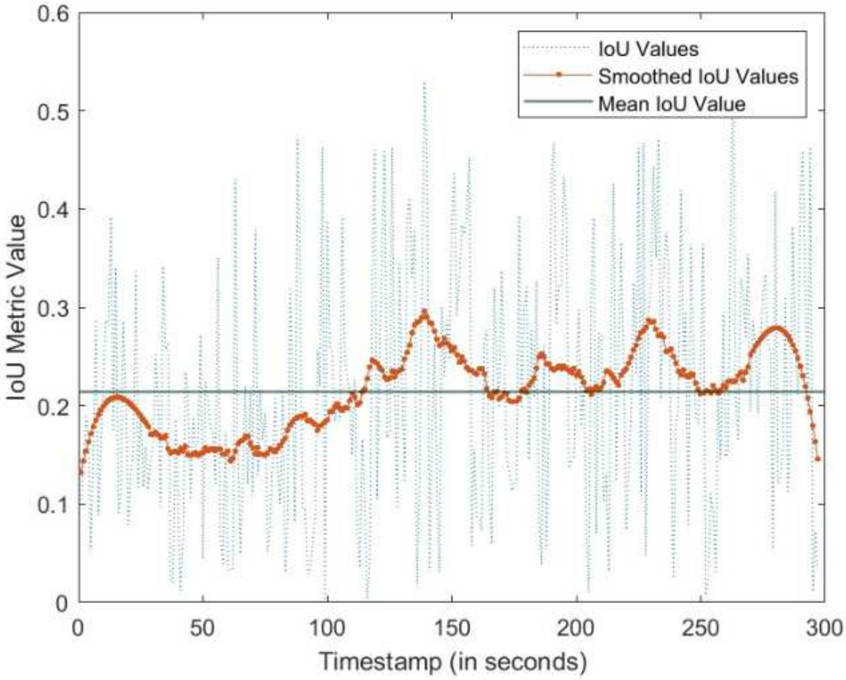

Figure 6.6: IOU graph for scene 4

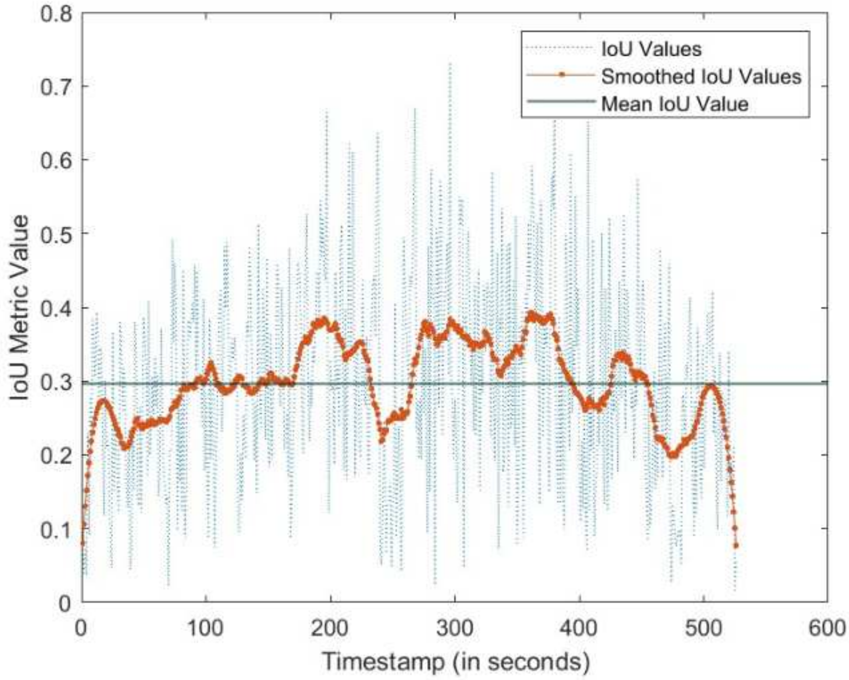

Figure 6.7: IOU graph for scene 5

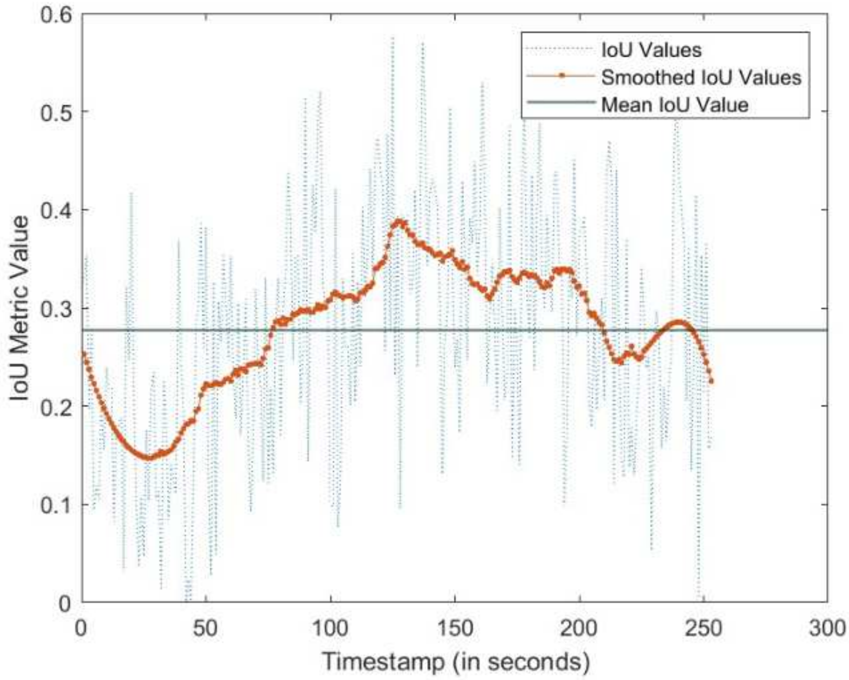

Figure 6.8: IOU graph for scene 6

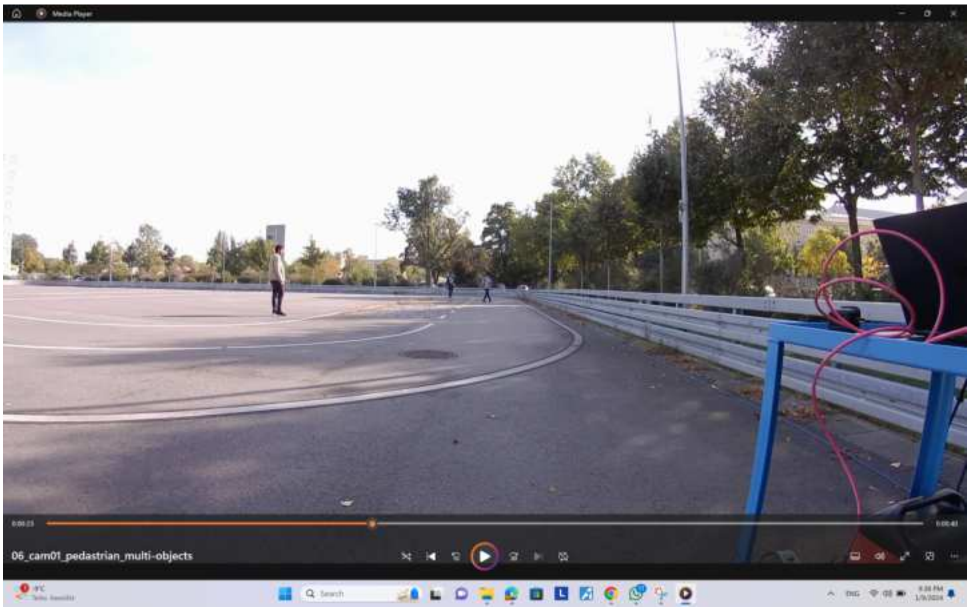

Figure 6.9: pedestrians

Figure 6.10: single car

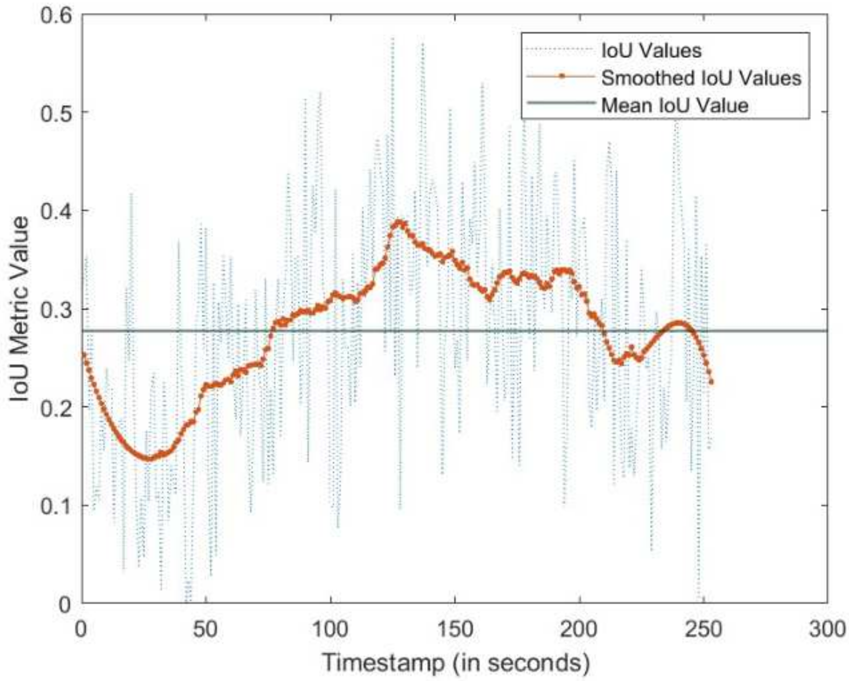

Figure 6.11: IOU graph for scene 6

6.3 Examples of labelling process

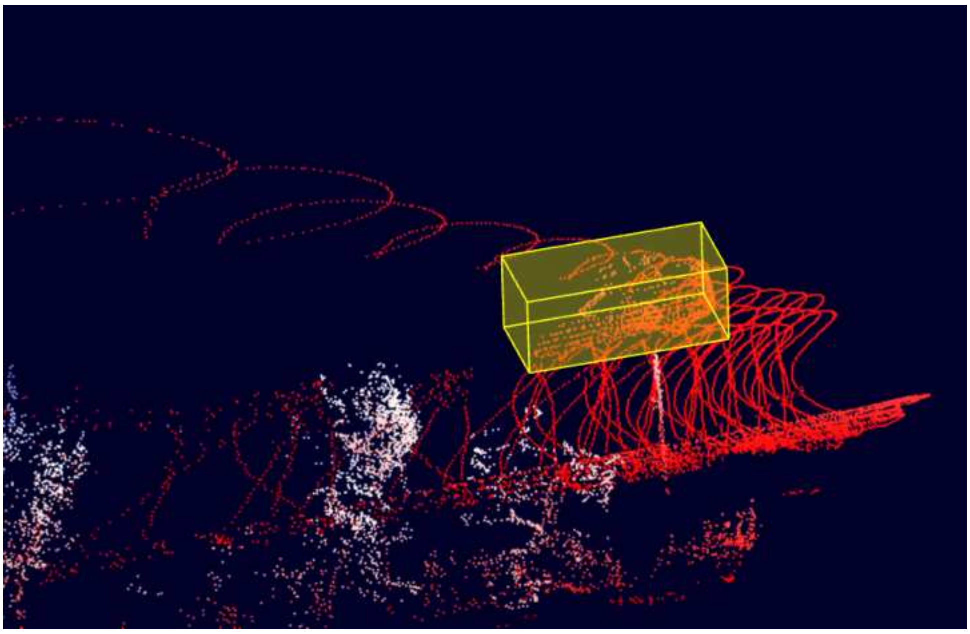

Figure 6.12: point clouds of a car is captured in this figure

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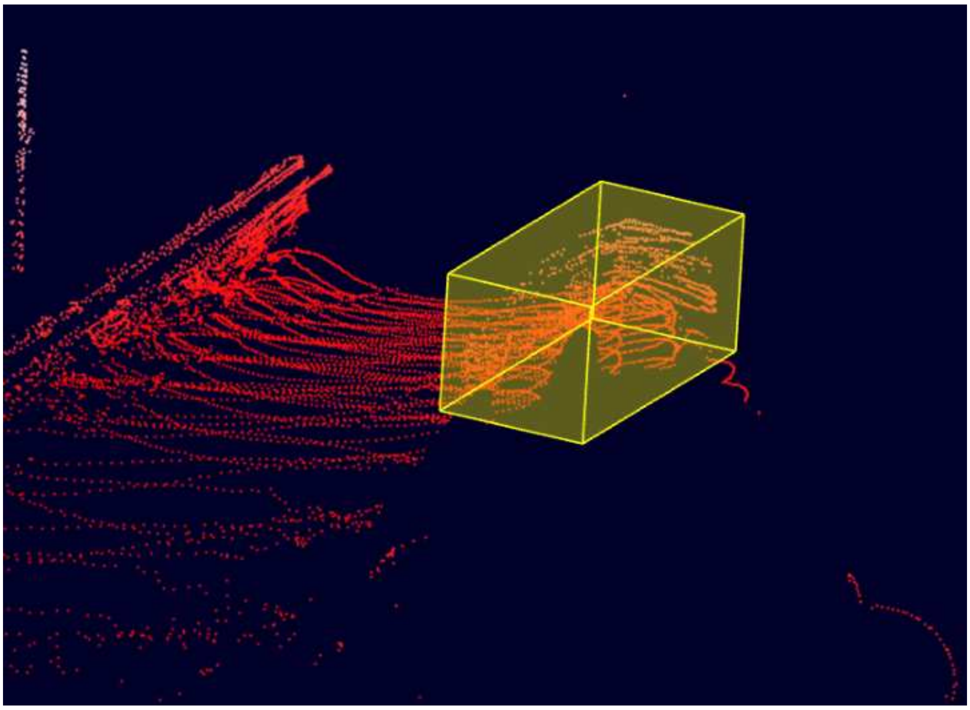

Figure 6.13: point clouds of a car is captured in this figure

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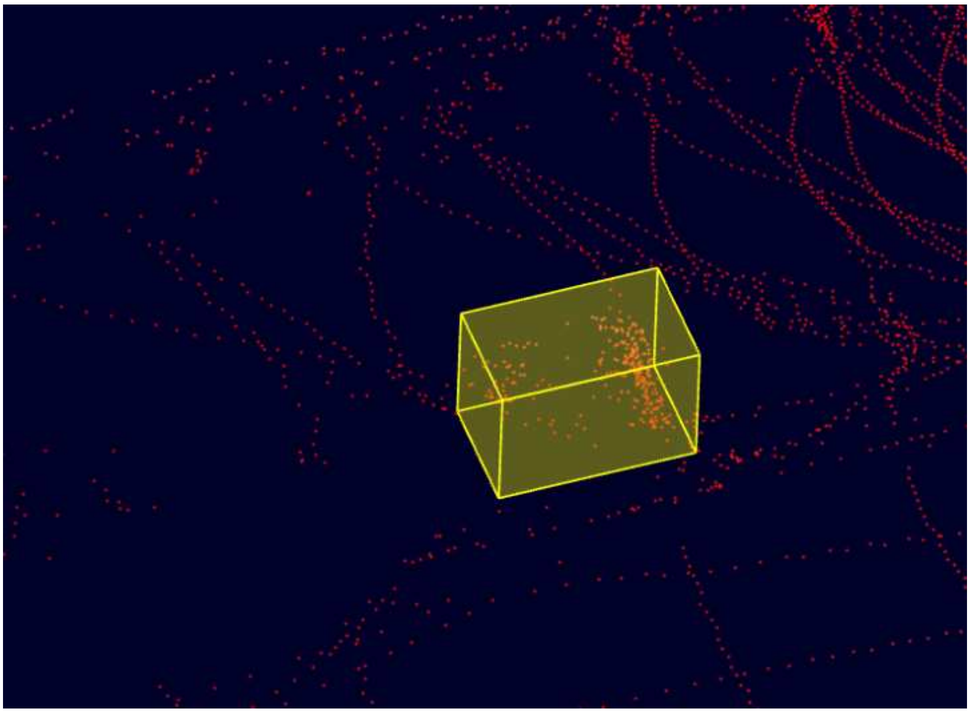

Figure 6.14: point clouds of a car is captured in this figure

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